



FIT2D: a multi-purpose data reduction, analysis and visualization program

A. P. Hammersley*

ESRF, CS40220, 38043 Grenoble Cedex 9, France. *Correspondence e-mail: hammersley@esrf.eu

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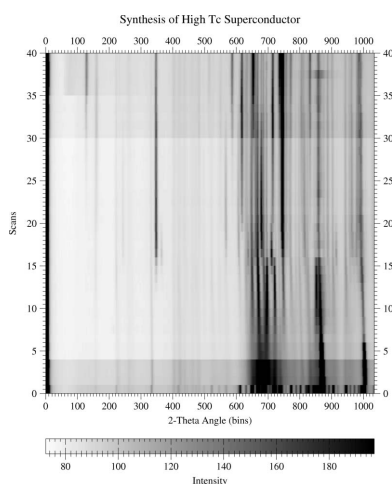
FIT2D is one of the principal area detector data reduction, analysis and visualization programs used at the European Synchrotron Radiation Facility and is also used by more than 400 research groups worldwide, including many other synchrotron radiation facilities. It has been developed for X-ray science, but is applicable to other structural techniques and is used in analysing electron diffraction data and microscopy, and neutron diffraction and scattering data. *FIT2D* works for both interactive and ‘batch’-style data processing. Calibration and correction of detector distortions, integration of two-dimensional data to a variety of one-dimensional scans, and one- and two-dimensional model fitting are the main uses. Many other general-purpose image processing and image visualization operations are available. Commands are available through a ‘graphical user interface’ and operations common to certain types of analysis are grouped within ‘interfaces’. Executable versions for most workstation and personal computer systems, and web page documentation, are available at <http://www.esrf.eu/computing/scientific/FIT2D>.

1. Introduction

FIT2D is a data reduction and analysis program developed at the European Synchrotron Radiation Facility (ESRF), to fulfil a variety of needs in different areas of scientific research. It has been developed through close collaboration with beamline scientists and users, and in conjunction with experimental and calibration methods. The user interface is graphical and was designed to be modular and aims to be easy to use. *FIT2D* is mainly written in Fortran90, with small sections written in ANSI-C.

For versatility, operations are modular and independent, *i.e.* the user may choose to perform operations in almost any order. A number of input file formats are available, including generic input options for unknown formats. For fast calculation data are stored in internal memory. Arrays to store estimated variance values may be created, and error propagation performed assuming Gaussian error distributions (Bevington & Robinson, 1992).

FIT2D uses two data arrays to store data and allow data manipulation operations. These are referred to as the ‘main data array’ and as the ‘memory’. The ‘memory’ is like the ‘memory’ function on many electronic calculators, only it stores a whole data array, auxiliary information and perhaps a variance array. The ‘main data array’ is where all input/output and display operations take place, and the ‘memory’ is where the output of non-trivial operations is placed. Within the graphical user interface, the ‘main data array’ and the ‘memory’ are swapped automatically so that the output is visible, and old data remain in the ‘memory’. The very widely available ‘EXCHANGE’ command allows the two data sets to be swapped efficiently.



Input image data are converted and stored in 32 bit floating-point reals, allowing non-integer and non-uniform operations and corrections, such as flat-field, dead-time and geometrical corrections. IEEE 32 bit floating-point reals allow values up to 10^{38} to be stored and with a precision of seven decimal digits. This is more than enough for any real experimental image data. Raw experimental image data involve errors which are at best Poissonian. Pixel values would need to be of the order of 10^{14} counts or more to warrant the use of 64 bit reals. (Note: certain operations such as matrix inversion can be sensitive to rounding errors, so internally they are carried out using 64 bit arithmetic.) Even on 64 bit machines there is a large efficiency gain by using 32 bit reals as opposed to 64 bit reals, as the data flow to and from RAM and disk is halved, and arithmetic operations are still significantly faster.

FIT2D, to a certain extent, is 'filling gaps' in the data analysis process relevant to ESRF users and other experimentalists, e.g. area detector calibration and integration of two-dimensional images to one-dimensional scans. It is designed to be used together with existing programs, e.g. for input into Rietveld programs, and *GSAS* (Larson & Von Dreele, 1994) and 'Powder Diffraction Standard' (*PDS*) (Murry & Fitch, 1990) formats may be created. Support of the 'BSL/OTOKO' file format was added to allow file interchange with the Hamburg/Daresbury small-angle scattering programs *OTOKO*, *BSL* and *KOHALA* (Boulin *et al.*, 1986; Mant, 1998), the *CCP13* software suite (<http://www.ccp13.ac.uk>) (Langan *et al.*, 1996; Hudson *et al.*, 1997), and other programs for small- and wide-angle scattering and fibre diffraction. Old formats continue to be supported to allow re-analysis of old data and to maintain interoperability. Recently, support for inputting Dectris Pilatus images has been added. ASCII and raw binary output allow generalized transfer to many other programs. PostScript and TIFF output formats allow graphics to be easily transferred to presentation programs. Further output formats can be added where there is a genuine demand.

There are four types of documentation: internal 'context-related' help pages, which should be sufficient for basic use; an introductory guide explaining user interaction and giving an overview of the available functions (Hammersley, 1997); a reference manual with detailed command descriptions (Hammersley, 1998a); and published articles explaining the algorithms used. Executable files, installation instructions, 'frequently answered questions' and *latex2html*-converted manuals (Drakos & Moore, 1997) are available from the *FIT2D* home page: <http://www.esrf.eu/computing/scientific/FIT2D>.

Two examples of use will be presented which demonstrate many of the capabilities: (i) calibrating the experimental geometry using a standard calibrant powder sample; (ii) a high-pressure kinematic study on the formation of high- T_c superconducting compounds.

2. General discussion of model fitting within *FIT2D*

Model fitting is used in many places within *FIT2D*, both explicitly and implicitly. For example, model fitting is used to

fit the diffraction cones produced by a NIST calibration powder standard to intensities found on a potentially non-orthogonal detector.

The model fitting uses well known algorithms such as those described by Gill *et al.* (1981). Implementation details have been vital in producing a robust and efficient system (Hammersley, 1998b), i.e.

(i) Graphical display is used to obtain good initialization and to allow inspection of results, to avoid problems of 'local minima'.

(ii) A scaling transformation is applied to the fit parameters to ensure that the fit function is roughly equally sensitive to changes in all the parameter values. This improves the rate of convergence and the robustness of the fitting.

(iii) Least-squares fitting is used; when data values are obtained with known Gaussian error distributions the least-squares method is strictly equivalent to the maximum likelihood method. The error estimates are often not Gaussian, but more seriously they are invariably unknown and poorly estimated (Schwarzenbach *et al.*, 1989). For counting statistics data this is more important than the difference between the Poissonian and Gaussian error distributions, even for low-count data. A simple method for largely reducing the introduced bias has been demonstrated and is implemented (Hammersley & Antoniadis, 1997).

3. The detector calibration and correction

The importance of detector distortion calibration is frequently underestimated. Sophisticated data analysis methods often give no significant advantage over more naive methods as both make false assumptions on the raw data. To put this point in a precise context: the most important factor in improving and extending multi-wavelength anomalous dispersion phasing techniques is to improve the accuracy of flat-field calibration of area detectors (Otwinowski, 1998).

3.1. Intensity linearity and background subtraction

All other corrections depend on the intensity values being on a linear scale. This is particularly true for accurate flat-field correction. Intensity linearity calibration data may be fitted using a polynomial of user-specified order and a look-up table is produced to allow fast image correction.

Background subtraction is very important; in removing non-X-ray signal background, e.g. dark counts in a CCD detector system, the remaining signal is put on a linear scale. This is a vital step in flat-field calibration.

3.2. Spatial distortion calibration and correction

Spatial distortion and calibration methods are well developed (Stanton *et al.*, 1992; Thomas, 1989; Svensson *et al.*, 1993). Calibration grid images are used to define the spatial distortion at discrete grid points, which are interpolated using a bi-cubic spline function fit, to a desired root-mean-square (RMS) accuracy (Dierckx, 1993). Fitting is important, since individual

grid positions have some error and holes can sometimes get partially blocked by small particles.

Two intensity-conserving methods are available to re-bin to an ideal grid. The first follows the method of Stanton *et al.* (1992) and makes the approximation that the edges of the distorted and corrected pixels are parallel. This is fast and works well for detectors where the distortions in the X and Y directions are nearly perpendicular, *e.g.* flat-bed image plate (IP) scanners. Unfortunately the assumption of parallel edges leads to a highly uneven redistribution of intensity when the distorted pixels are highly rotated. Alternatively the corners of distorted pixels are calculated on the output grid, and the intensity is distributed proportionally to the covered area. This involves much more calculation and is much slower. Tests with single-crystal macromolecular diffraction images and integration by *DENZO* (Otwinowski, 1993) show that the improvement in data quality is significant, especially for the high-resolution data.

For faster correction, a look-up table of pixel shifts, and the relative contributions of input pixels to output pixels, can be calculated. The look-up table is very large and for fast correction must be stored in computer RAM. The accuracy of the look-up table is important. If one byte is used to store the contributions then there is a rounding error of up to $(0.5/255)$ for each re-binning operation. Each output pixel has contributions from typically four input pixels, but the errors are highly correlated, producing an aliasing effect. In a real case of an X-ray image intensifier detector, the RMS error was found to be 0.23% (Hammersley *et al.*, 1994). Initial tests indicate that reduction in data quality, as measured by R_{merge} statistics, is insignificant, suggesting that this approximation is acceptable.

3.3. Flat-field calibration and correction

Accuracy in flat-field correction is fundamental to the accuracy in corrected data; for example, a low-frequency 1% inaccuracy in the correction will affect Bragg peaks to the same level.

Whilst theoretically simple, accurate flat-field calibration is difficult in practice (Thomas, 1990; Hammersley *et al.*, 1995). Simple calibration and correction depend on an isotropic point source at the energy being calibrated. Fluorescence from doping agents suspended in light-element amorphous glass matrices has been used to obtain radiation close to isotropic, but several per cent variation makes it necessary to calibrate the calibrant (Moy *et al.*, 1996). The main implemented method is to use an emission which is as smoothly changing and rotationally symmetric as possible about the direct beam. This may be characterized with great accuracy using a 2θ scan with a photon-counting detector (Hammersley *et al.*, 1995).

2θ -scan data may be corrected for beam decay and detector dead time, and interpolated using a one-dimensional polynomial of user-specified order. The interpolation function is used to correct a 'flood-field' image of the fluorescent source to an equivalent 'flat-field' image. This corrects for anisotropy, the difference between the constant-distance 2θ scan and the

flat-plate geometry, and if required the additional air absorption. The resulting image can be normalized, and the reciprocal taken and used for flat-field correction by pixel-by-pixel multiplication.

It is possible to use another area detector as a reference (Stanton *et al.*, 1992). Invariably this method requires that one of the images be re-binned and aligned to the other image pixel grid, so that the two are directly comparable. This technique is limited, at best, to the accuracy of the reference detector.

3.4. Image-plate intensity signal decay

The decay of the latent image signal with time has been studied by various groups (Hall *et al.*, 1991; Fujii *et al.*, 1991; Templar, 1991; Shaw *et al.*, 1992). Signal decay during scanning results in a change of uniformity of intensity response which can be significant when scanning times are long. For single-crystal diffraction using spiral read-out scanners centred on the direct beam, all symmetry-related Bragg peaks are scanned together so R_{merge} is not significantly affected. The decay results in an incorrect temperature factor so is usually not a significant problem. However, for flat-bed scanners this change can degrade results.

Correction is possible if the decay of the particular type of IP material has been calibrated, and if the times at the start and the end of the exposure, and at the start and end of the scan, are noted. The decay is modelled as the sum of two exponential decays (Hammersley *et al.*, 1995), integrated analytically, and a correction factor is calculated and applied. This can be a 5–10% effect, reduced to 0.3% with correction.

4. General image processing operations

The Image Processing (General) interface contains commands for normal mathematical operations, geometrical transformations, and filtering of one-dimensional and two-dimensional data. An example which demonstrates the use of the 'memory' is the subtraction of one image from another, *e.g.* subtracting a dark-current image from a diffraction image. The following commands are used:

INPUT: input dark-current image.

EXCHANGE: place the dark-current image in the 'memory'.

INPUT: input diffraction image.

MATHS: enter Maths sub-menu.

SUBTRACT: subtract the 'memory' image from the current data array image.

EXIT: exit Maths sub-menu.

OUTPUT: save dark-current-subtracted data in a file.

(The 'memory' still contains the dark-current image, so another diffraction image could be input without having to re-input the dark current.)

5. User one-dimensional and two-dimensional model fitting

One-dimensional fit models may be defined using direct coordinate input with the raw data, constructed from an

arbitrary number of peak functions (Gaussian, Lorentzian and Voigt), trigonometric or exponential decay functions, and a polynomial function of specified order (Hammersley & Riekkel, 1989). The Voigt function allows the natural line width (Lorentzian) to be fitted convolved with other factors, mainly the instrument response, approximated by a Gaussian profile. Setup options allow control parameters to be varied, *e.g.* weighted fitting or not (provided variance arrays exist), and individual model parameters to be constrained, or freely vary. The fitted model and the data are shown in the upper graph in Fig. 1, and beneath are the residuals.

A two-dimensional fit model can be composed of a user-selected mixture of two-dimensional Gaussians, 'polar Gaussians', 'row-lines' and a two-dimensional polynomial function. A 'polar Gaussian' is a two-dimensional function which has a Gaussian profile in a radial sense, and in an azimuthal sense, based on a symmetry centre point. A 'row-line' function is a series of 'polar Gaussians' simulating the broad Bragg peaks produced by partially disordered fibre samples, *e.g.* the fitting of collagen row-lines (Wess *et al.*, 1998).

6. Online display and data quality monitoring

As well as 'offline' data analysis, 'online' input and display is available. Files are automatically input and an 'alarm' may be set to warn if a file does not appear within a set time. A peak search algorithm can find diffraction peaks and produce rough estimates of signal-to-noise ratios as a function of resolution. This is approximate and is intended to give an indication of data quality during data collection.

7. Powder diffraction and small- and wide-angle scattering with area detectors

For powder diffraction, area detectors are complementary to traditional high-resolution powder diffractometers, producing lower-resolution data, but gaining enormously in collection efficiency. With large IP systems the averaging for the high-resolution data may be as large as 10 000. This gain together with the two-dimensional information gives the following main applications:

Small- and wide-angle scattering. The same type of angular integration, but with the output bins being defined in Q space.

Small sample volumes and/or weakly scattering samples. High pressure: diamond anvil cells lead to very small sample volumes and high backgrounds (Piltz *et al.*, 1992; Hammersley *et al.*, 1996). Microfocus experiments: high-brilliance synchrotrons with newly developed focusing techniques allow illumi-

nation and scanning of sample areas of the order of a micrometre (Bul  on *et al.*, 1997). Microwires: intrinsically small samples. Fullerenes: the resolution is well matched to such intermediate crystallinity samples. Dilute samples: samples may be naturally dilute or deliberately diluted to minimize local order.

Subtle effects. Some phase changes may only be recognized by tiny changes in the diffraction pattern. The time taken to record data may be prohibitive on a high-resolution diffractometer.

Time-resolved studies. With an online detector, good statistics can be obtained in a very short time.

Sample orientation and order/disorder. Sample orientation and degree of order can be studied from the azimuthal information.

Texture analysis. Azimuthal intensity profiles give information on sample texture.

Sample strain. Under extreme conditions the d spacings can change as a function of azimuth.

Beamline calibration. See §9.

The Powder Diffraction (2D) interface allows Debye–Scherrer rings to be integrated to the equivalent of a 2θ scan with a zero-dimensional detector (Hammersley *et al.*, 1996). Rings may be fitted with ellipses to determine the beam centre and the non-orthogonality of the detector to the direct beam. (Spatial distortion in the detector must be corrected beforehand. The effects are often small, and uncorrected spatial distortion can lead to incorrect and unstable results.) It is important to be able to mask out bad pixels during integration. As well as detector defects, bad pixels and bad regions may result from shadowing owing to the experimental setup or diffraction peaks from contaminants within the beam. Arbitrary polygon mask regions and different-sized circular

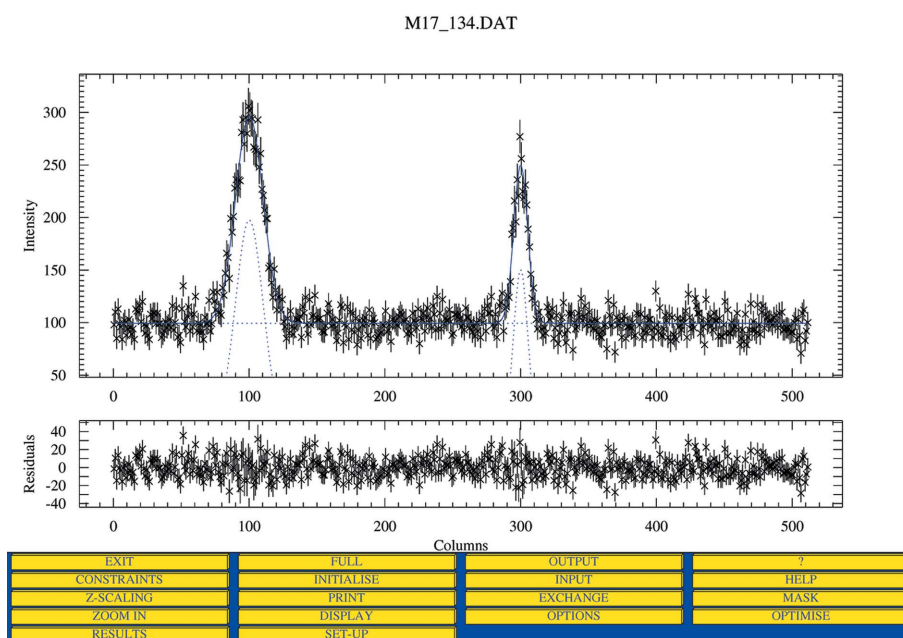


Figure 1
Example of the MFIT interface menu and display.

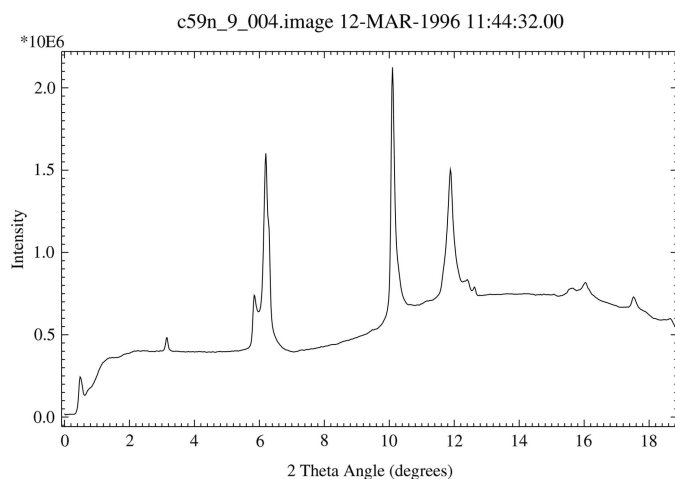


Figure 2
2 θ scan of (C₅₉N)₂ integrated by *FIT2D* from an image recorded on a 30 cm MarResearch scanner.

regions may be defined graphically, and an upper or lower threshold value may be used to define a mask automatically.

Account is taken of the difference in geometry, detector non-orthogonality and polarization effects as a function of both 2 θ angle and azimuth (Kahn *et al.*, 1982). Flexibility is provided by allowing the integration region to be defined graphically, and allowing a choice of the number of output bins in both the azimuthal and the 2 θ direction. In the 2 θ sense, a choice of equal angle bins, equal radial distance bins, equal Q -space bins or equal d -spacing bins is available, and a variety of versatile one-dimensional and two-dimensional 'polar transformations' are available, *e.g.* a number of different 2 θ scans for different azimuth ranges, a one-dimensional profile of intensity of a ring as a function of azimuth for texture studies. Output formats include *GSAS* (Larson & Von Dreele, 1994), *PDS* (Murry & Fitch, 1990), *Cerius*¹ and a general-purpose ASCII two-column format, for input into spreadsheet and other programs.

An example is the study of azafullerene (C₅₉N)₂ by Brown *et al.* (1996) and Kordatos *et al.* (1998). The integrated spectrum obtained from an azafullerene sample is shown in Fig. 2. Data were collected on the Swiss–Norwegian beamline at the ESRF using a wavelength of 0.8736 Å and a 30 cm MarResearch online IP scanner. The background is relatively high, which is normal for this type of sample; the high averaging leads to good statistics in both background and peak values. The broad peak shape is intrinsic to the sample and much larger than the resolution of the technique.

8. Macros and file series processing

Often the same or similar data analysis sequences must be repeated on a file series. A file series is a sequence of related images stored with the same data format in different files. File names of such series have a numerical part which changes in a regular manner, *e.g.* file_001.dat, file_002.dat, ...,

file_365.dat, or sas1, sas2, ..., sas99. The File Series interface contains commands to average, display and integrate series of files. An example of use of the INTEGRATE command is given in §10.

Analysis sequences that are not available as FILE SERIES commands can be repeated using macros. All commands including the graphical user interface commands can be accessed within macros and loops may be defined. Interactive input is also available, and variables may be defined and their values used to build file names *etc.*

9. Using a powder calibrant sample to calibrate the experimental geometry

Given a high-quality image of a calibrant powder with accurately known d spacings, it is possible to calibrate not only the beam centre and detector non-orthogonality, but also the sample-to-detector distance or X-ray wavelength. If the data statistics are good, at least two rings are available, and especially if high-angle rings are present, then it is possible to refine all parameters including the sample distance and wavelength simultaneously with accurate results.

This is possible because the effect of wavelength and distance are slightly different. If s is the sample-to-detector distance, λ is the wavelength and d is the d spacing of the reflection, then the radial distance r of the reflection on a detector orthogonal to the direct beam is

$$r = s \tan[2 \arcsin(\lambda/2d)]. \quad (1)$$

For a detector which is tilted by an angle φ away from the orthogonal to the direct beam, the equation of the ellipse as measured on the tilted detector of the reflection in the plane of the tilt is

$$\cos^2 \varphi x^2 + y^2 = (s + \sin \varphi x)^2 \tan^2[2 \arcsin(\lambda/2d)]. \quad (2)$$

From these equations all the parameters may be refined. An example of the output is shown in Fig. 3. The data points used to fit the geometry parameters are shown as crosses, together with the predicted rings.

A choice of six calibrants is available: alumina, cerium dioxide, paraffin wax, lanthanum hexaboride, silicon and sodium chloride. Lanthanum hexaboride is probably the best choice owing to its minimal graininess. Paraffin wax is a poor calibrant, but is nevertheless included since it is often used, and whilst the results will be less accurate, they are usually sufficient for biological samples whose cell sizes often show variation. Evidently wax samples cannot be used to refine accurately both wavelength and sample distance. This technique has been verified to agree to within the accuracy obtained scanning an absorption edge using a Ge detector on ID-30 at the ESRF (Kunz, 1997). An A3-sized IP was used to record a silicon powder pattern and scanned using a Molecular Dynamics Storm IP scanner. This scanner has slight spatial distortion and tests showed this degrades integrated scan quality if not corrected. Since the spatial distortion is stable it can be reliably corrected. Results from an absorption-edge

¹ Crystallographic software package from Molecular Simulations.

Table 1

X-ray wavelength calibration by scanning an absorption edge and by using a silicon powder pattern.

Edge scan wavelength (Å)	0.4023	0.4254	0.40945
<i>FIT2D</i> refined value (Å)	0.4022	0.4254	0.40940

scan in the vicinity of the wavelength used and corresponding results from silicon powder pattern data are shown in Table 1.

10. Analysis of high-pressure kinematic data

The FILE SERIES INTEGRATE command allows integration of two-dimensional data to one-dimensional scans as described in §7. Once the integration of one image has been specified interactively, the integration is repeated automatically for the whole series. There are options to correct detector distortions prior to integration and to output individual integrated scans. The series of one-dimensional scans are stored as a two-dimensional image, showing trends in d spacings and phase changes.

This technique has been exploited by Toulemonde *et al.* (2000) in a high-pressure kinematic study on ID-30 at the ESRF. A large-volume Paris–Edinburgh high-pressure cell

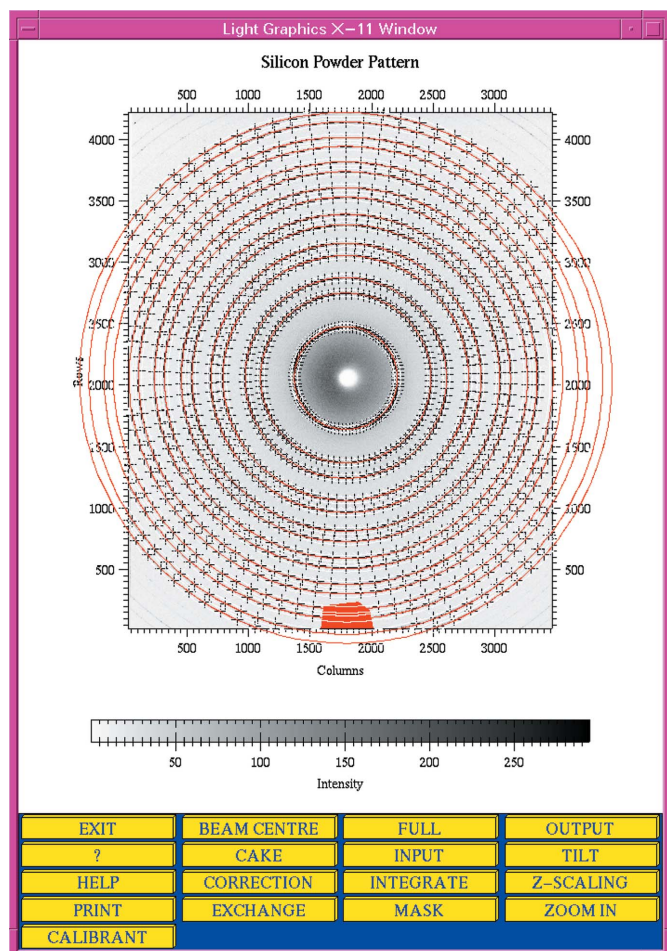
was used to investigate the formation of high- T_c superconducting compounds. A fast read-out online IP detector (Thoms *et al.*, 1998) was used to collect *in situ* X-ray powder diffraction data during the process of synthesis at high pressure and high temperature.

A series of one-dimensional integrated 2θ scans are shown in Fig. 4, corresponding to the initial products (first line), the initial products at high pressure (second–fourth lines), and subsequently the heating to high temperature and the changes due to synthesis with time. This diagram and similar ones show clearly changes in d spacings, composition and phase owing to application of pressure and temperature.

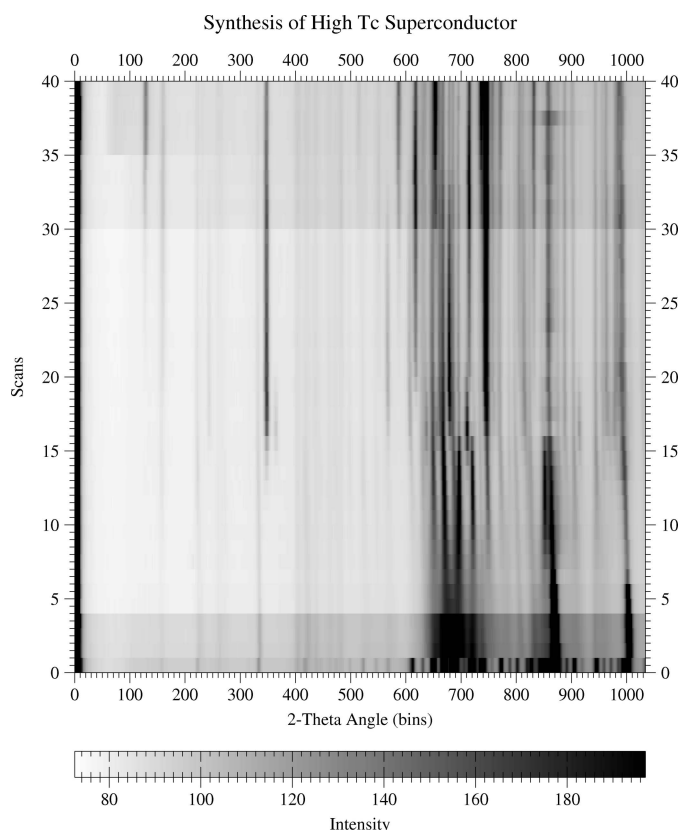
Versatile and efficient analysis of enormous quantities of data (gigabytes or terabytes) is possible. Even with a non-optimum computing environment, the data analysis can keep pace with data collection. With a dedicated computer system specifically optimized for this type of data analysis based on cheap PC hardware (*e.g.* 3.4 GHz processor, 4 Gbytes RAM, medium-fast hard disk) it should be possible to integrate 53 Gbytes of raw data per hour (1.25 Tbytes per day), including correction of detector distortions.

11. Availability

FIT2D is available free of charge for academic research in return for an undertaking for appropriate acknowledgement. Commercial users are requested to pay a yearly fee (presently 2500 euros). Executables for various machines and operating

**Figure 3**

Silicon powder pattern used to refine beam centre, detector tilt, sample-to-detector distance and X-ray wavelength.

**Figure 4**

Series of 2θ scans taken during the synthesis of a high- T_c superconductor.

systems can be obtained from the web. Versions are available for Windows 7 (and newer), Linux and Mac OSX. Older versions are available for Windows XP, NT, 98 and a variety of Unix systems.

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References

- Bevington, P. R. & Robinson, D. K. (1992). *Data Reduction and Error Analysis for the Physical Sciences*, 2nd ed. New York: McGraw-Hill, Inc.
- Boulin, C., Kempf, R., Koch, M. H. J. & McLaughlin, S. M. F. (1986). *Nucl. Instrum. Methods Phys. Res. Sect. A*, **249**, 399–407.
- Brown, C. M., Cristofolini, L., Kordatos, K., Prassides, K., Bellavia, C., González, R., Keshavarz-K., M., Wudl, F., Cheetham, A. K., Zhang, J. P., Andreoni, W., Curioni, A., Fitch, A. N. & Pattison, P. (1996). *Chem. Mater.* **8**, 2548–2550.
- Bulón, A., Pontoire, B., Riekkel, C., Chanzy, H., Helbert, W. & Vuong, R. (1997). *Macromolecules*, **30**, 3952–3954.
- Dierckx, P. (1993). *Curve and Surface Fitting with Splines*. Oxford University Press.
- Drakos, N. & Moore, R. (1997). *The LATEX2HTML Translator*, <http://www.eng.fsu.edu/~dommelen/l2h/latex2html.pdf>.
- Fujii, I., Morimoto, Y., Higuchi, Y., Yasuoka, N., Katayama, C. & Miki, K. (1991). *Acta Cryst.* **B47**, 137–144.
- Gill, P. E., Murray, W. & Wright, M. H. (1981). *Practical Optimization*. London: Academic Press.
- Hall, C. J., Lewis, R. A., Parker, B. & Worgan, J. (1991). *Nucl. Instrum. Methods Phys. Res. Sect. A*, **310**, 215–219.
- Hammersley, A. P. (1997). *FIT2D: an Introduction and Overview*. Internal Report ESRF97HA02T, ESRF, Grenoble, France.
- Hammersley, A. P. (1998a). *FIT2D*. Version 9.129. Reference Manual V3.1. Internal Report ESRF98HA01T, ESRF, Grenoble, France.
- Hammersley, A. P. (1998b). *Model Fitting Within FIT2D*, <http://www.esrf.eu/computing/scientific/FIT2D/MF/model.html>.
- Hammersley, A. P. & Antoniadis, A. (1997). *Nucl. Instrum. Methods Phys. Res. Sect. A*, **394**, 219–224.
- Hammersley, A. P. & Riekkel, C. (1989). *Syn. Rad. News*, **2**, 24–26.
- Hammersley, A. P., Svensson, S. O., Hanfland, M., Fitch, A. N. & Häusermann, D. (1996). *High Press. Res.* **14**, 235–248.
- Hammersley, A. P., Svensson, S. O. & Thompson, A. (1994). *Nucl. Instrum. Methods Phys. Res. Sect. A*, **346**, 312–321.
- Hammersley, A. P., Svensson, S. O., Thompson, A., Graafsma, H., Kvick, Å. & Moy, J. P. (1995). *Rev. Sci. Instrum.* **66**, 2729–2733.
- Hudson, L., Harford, J. J., Denny, R. C. & Squire, J. M. (1997). *J. Mol. Biol.* **273**, 440–455.
- Kahn, R., Fourme, R., Gadet, A., Janin, J., Dumas, C. & André, D. (1982). *J. Appl. Cryst.* **15**, 330–337.
- Kordatos, K., Brown, C. M., Prassides, K., Bellavia-Lund, C., de la Cruz, P., Wudl, F., Thompson, J. D. & Fitch, A. N. (1998). *Molecular Nanostructures*, edited by H. Kuzmany, J. Fink, M. Mehring & G. Roth. Singapore: World Scientific.
- Kunz, M. (1997). Personal communication.
- Langan, P., Denny, R. C., Mahendrasingam, A., Mason, S. A. & Jaber, A. (1996). *J. Appl. Cryst.* **29**, 383–389.
- Larson, A. C. & Von Dreele, R. B. (1994). *General Structure Analysis System (GSAS) Manual*. LANSCE, MS-H805, Los Alamos National Laboratory, Los Alamos, NM, USA.
- Mant, G. R. (1998). *SRS Program Library*, <http://www.dl.ac.uk/SRS/NCD>.
- Moy, J. P., Hammersley, A. P., Svensson, S. O., Thompson, A., Brown, K., Claustre, L., Gonzalez, A. & McSweeney, S. (1996). *J. Synchrotron Rad.* **3**, 1–5.
- Murry, A. D. & Fitch, A. N. (1990). *Powder Diffraction Program Library (PDPL)*. University College London, UK.
- Otwinowski, Z. (1993). Oscillation Data Reduction Program. *Proceedings of the CCP4 Study Weekend: Data Collection and Processing*, 29–30 January 1993, compiled by L. Sawyer, N. Isaacs & S. Bailey, pp. 56–62. SERC Daresbury Laboratory, England.
- Otwinowski, Z. (1998). Personal communication.
- Piltz, R. O., McMahon, M. I., Crain, J., Hatton, P. D., Nelves, R. J., Cernik, R. J. & Bushnell-Wye, G. (1992). *Rev. Sci. Instrum.* **63**, 700–703.
- Schwarzenbach, D., Abrahams, S. C., Flack, H. D., Gonschorek, W., Hahn, Th., Huml, K., Marsh, R. E., Prince, E., Robertson, B. E., Rollett, J. S. & Wilson, A. J. C. (1989). *Acta Cryst.* **A45**, 63–75.
- Shaw, C. C., Herron, J. M. & Gur, D. (1992). *Proc. SPIE*, **1651**, 156–162.
- Stanton, M., Phillips, W. C., Li, Y. & Kalata, K. (1992). *J. Appl. Cryst.* **25**, 549–558.
- Svensson, S. O., Hammersley, A. P., Thompson, A., Gonzalez, A. & Ursby, T. (1993). *CCP4/EPS-EACBM Newsletter*, 29.
- Templer, R. H. (1991). *Nucl. Instrum. Methods Phys. Res. Sect. A*, **300**, 357–366.
- Thomas, D. J. (1989). *Proc. R. Soc. London Ser. A*, **425**, 129–167.
- Thomas, D. J. (1990). *Proc. R. Soc. London Ser. A*, **428**, 181–214.
- Thoms, M., Bauchau, S., Häusermann, D., Kunz, M., Bihan, T. L., Mezouar, M. & Strawbridge, D. (1998). *Nucl. Instrum. Methods Phys. Res. Sect. A*, **413**, 175–184.
- Toulemonde, P., Floch, S. L., Bordet, P., Capponi, J. J., Mezouar, M. & Odier, P. (2000). *Supercond. Sci. Technol.* **13**, 1129–1134.
- Wess, T. J., Hammersley, A. P., Wess, L. & Miller, A. (1998). *J. Mol. Biol.* **275**, 255–267.